

Airline Multi-Leg Seat Inventory & Overbooking Engine

Context: An airline client sells seats across multi-leg itineraries where a single physical seat inventory is shared across overlapping flight segments, and controlled overbooking is a deliberate revenue strategy.

Scope: Build a booking-inventory service for a small simulated route network (a handful of cities, connecting flights) that tracks seat availability per leg, allows a configurable overbooking factor per flight, and handles cancellations/rebookings that ripple across itineraries. You do not need a real GDS integration - a simulated flight schedule and fare classes are enough. Focus on the shared-inventory-across-itineraries model and the deliberate overbooking policy, not on fare pricing sophistication.

Requirements: Route/flight inventory model, itinerary booking API (single or multi-leg), configurable overbooking limit per flight, cancellation/rebooking flow, a "bumped passenger" resolution path when a flight is oversold at departure.

Constraints: A single seat sold on a multi-leg itinerary consumes inventory on every leg it touches, and legs are shared across many different itineraries. Overbooking must be a tunable, per-flight, non-hardcoded parameter, not a fixed global buffer. Two concurrent bookings for the last seat on a shared leg must not both succeed inconsistently across the itineraries that depend on it. Cancelling one leg of a connecting itinerary must not silently leave the other leg's inventory or the passenger's record in an inconsistent state.

Expectation : Shared-inventory data model across itineraries, concurrency-safe seat-claiming mechanism, overbooking-policy design and bump-resolution logic, explanation of what happens to dependent legs when one leg of an itinerary is cancelled or changed.

Demo requirement: Live demonstration of: (a) a last-seat concurrency race with

correct accept/reject counts, (b) a cross-itinerary shared-leg race with no oversell, (c) a cancellation cascading through a multi-leg itinerary with a shown final state, (d) an overbooking-limit change taking effect on in-flight bookings, and (e) a reconciliation check confirming leg inventory matches booking records after all of the above.